

Delayed anaphylaxis, angioedema, or urticaria after consumption of red meat in patients with IgE antibodies specific for galactose- α -1,3-galactose

Background Carbohydrate moieties are frequently encountered in food and can elicit IgE responses, the clinical significance of which has been unclear. Recent work, however, has shown that IgE antibodies to galactose- α -1,3-galactose (α -gal), a carbohydrate commonly expressed on nonprimate mammalian proteins, are capable of eliciting serious, even fatal, reactions. Objective We sought to determine whether IgE antibodies to α -gal are present in sera from patients who report anaphylaxis or urticaria after eating beef, pork, or lamb. Methods Detailed histories were taken from patients presenting to the University of Virginia Allergy Clinic. Skin prick tests (SPTs), intradermal skin tests, and serum IgE antibody analysis were performed for common indoor, outdoor, and food allergens. Results Twenty-four patients with IgE antibodies to α -gal were identified. These patients described a similar history of anaphylaxis or urticaria 3 to 6 hours after the ingestion of meat and reported fewer or no episodes when following an avoidance diet. SPTs to mammalian meat produced wheals of usually less than 4 mm, whereas intradermal or fresh-food SPTs provided larger and more consistent wheal responses. CAP-RAST testing revealed specific IgE antibodies to beef, pork, lamb, cow's milk, cat, and dog but not turkey, chicken, or fish. Absorption experiments indicated that this pattern of sensitivity was explained by an IgE antibody specific for α -gal. Conclusion We report a novel and severe food allergy related to IgE antibodies to the carbohydrate epitope α -gal. These patients experience delayed symptoms of anaphylaxis, angioedema, or urticaria associated with eating beef, pork, or lamb.